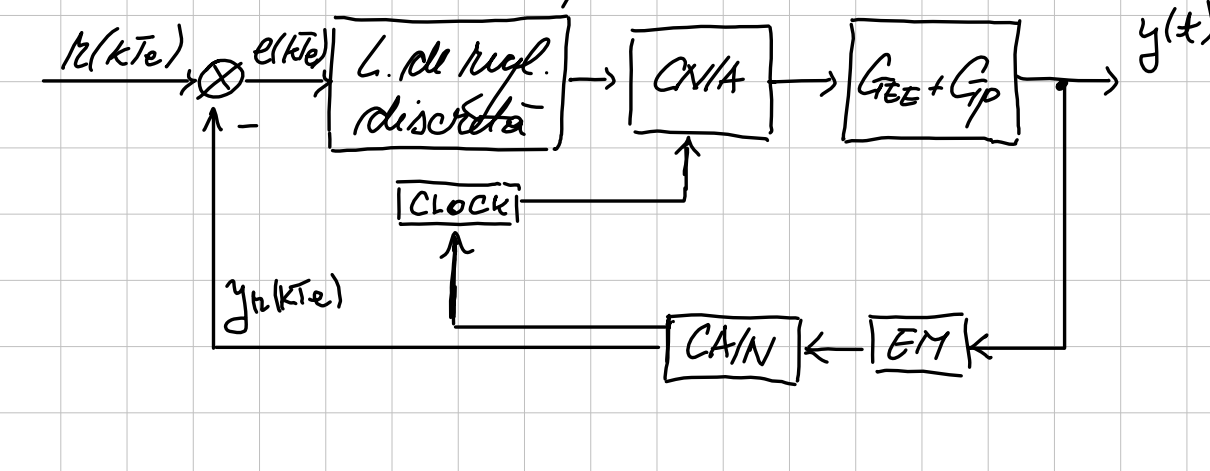
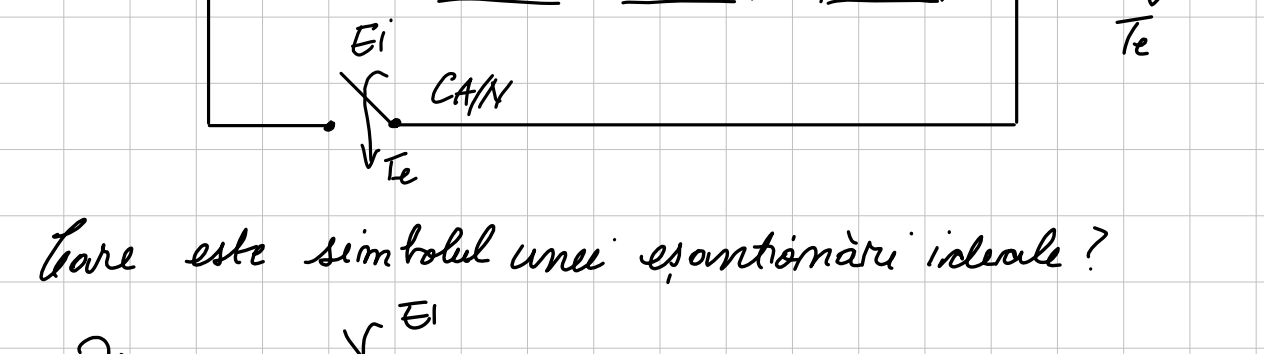
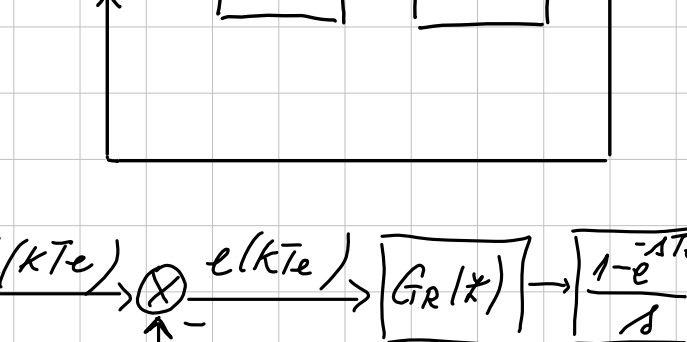


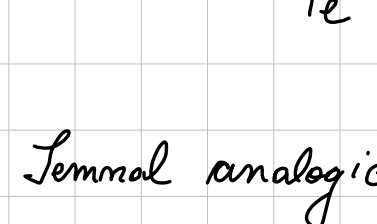
Recomputare



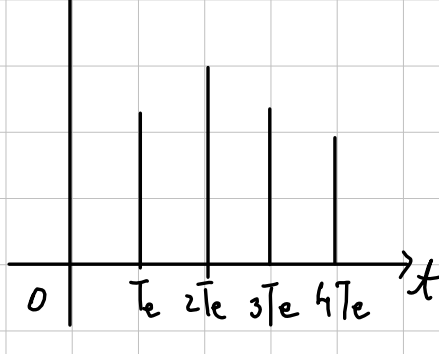
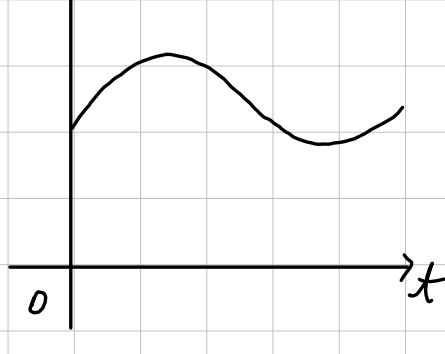
Sist. discret = are componente numerice



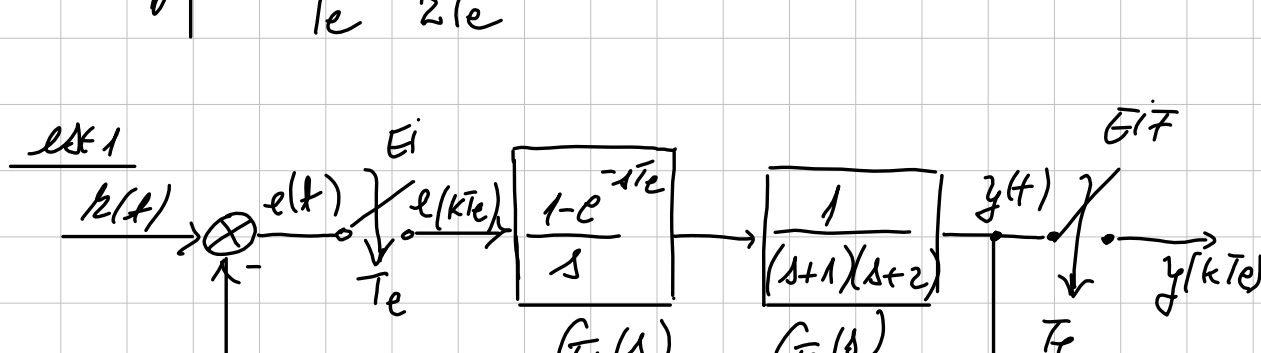
Care este simbolul unei esantionări ideale?



Semnal analogic



fenomenul aliasing

1) $G(z)$; $G_e(z)$ = ?2) e^*_{st} = ?

3) Stabilitate?

4) $y(kT_e)$ = ? (năp. sistemului) 5) spațiul stării lor

indic. de calitate? fot. de ff. a cccu dinde

MET I: $G_0(z) = \frac{\mathcal{Z}\{G_1(s) \cdot G_2(s)\}}{1 + \mathcal{Z}\{G_1(s) \cdot G_2(s)\}}$ ($G_0(z) = \frac{\mathcal{Z}\{G_1\} \cdot \mathcal{Z}\{G_2\}}{1 + \mathcal{Z}\{G_1\} \cdot \mathcal{Z}\{G_2\}}$ în alte cazuri)

fot. de ff. a buclii

 $G(z) = \mathcal{Z}\left\{\frac{1 - e^{-sT_e}}{s} \cdot \frac{1}{(s+1)(s+2)}\right\} = (1 - z^{-1}) \mathcal{Z}\left\{\frac{1}{s(s+1)(s+2)}\right\}$ $= \frac{(z-1)}{z} \mathcal{Z}\left\{\frac{C_1}{s} + \frac{C_2}{s+1} + \frac{C_3}{s+2}\right\}$ Ac. calc. C_1, C_2, C_3 $\Rightarrow \frac{(z-1)}{z} \mathcal{Z}\left\{\frac{1}{2} \cdot \frac{1}{s} - \frac{1}{s+1} + \frac{1}{2} \cdot \frac{1}{s+2}\right\}$ $= \frac{z-1}{z} \left\{0,5 \frac{z}{z-1} - \frac{z}{z-e^{-T_e}} + 0,5 \frac{z}{z-e^{-2T_e}}\right\} \quad T_e=1$ $= (z-1) \left(\frac{0,5(z-e^{-T_e})(z-e^{-2T_e}) - (z-1)(z-e^{-2T_e}) + 0,5(z-1)(z-e^{-T_e})}{(z-1)(z-e^{-T_e})(z-e^{-2T_e})} \right)$ $= (z-1) \left(\frac{0,5(z-e^{-1})(z-e^{-2}) - (z-1)(z-e^{-2}) + 0,5(z-1)(z-e^{-1})}{(z-1)(z-e^{-1})(z-e^{-2})} \right)$ $= \frac{0,7z + 0,07}{(z-0,36)(z-0,13)}$ MET II: $G(z) = G(s) \Big|_{s=\frac{z-1}{T_e}; \frac{z-1}{T_e}; \frac{2}{T_e} \cdot \frac{z-1}{z+1}}$ $\frac{1}{\left(\frac{z-1}{T_e} + 1\right)\left(\frac{z-1}{T_e} + 2\right)} \quad G(z) = \frac{kd(z+1)}{(z-e^{-T_e})(z-e^{-2T_e})} \quad \begin{matrix} m=2 \\ m=0 \\ m \neq m \\ (m-1)(z+1) \end{matrix}$ $\lim_{s \rightarrow 0} s^\alpha G(s) = \lim_{z \rightarrow 1} \left(\frac{z-1}{T_e z}\right)^\alpha \frac{1}{\left(\frac{z-1}{T_e} + 1\right)\left(\frac{z-1}{T_e} + 2\right)}$ avem 2 integroare $\Rightarrow \alpha = 0, 1, 2$

MET III: ec. cu derivate

 $G(s) = \frac{1}{s^2 + 3s + 2} = \frac{Y(s)}{U(s)} \Rightarrow s^2 Y(s) + 3s Y(s) + 2Y(s) = U(s) \Big|_{s^{-1}}$ $\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = u(t)$ $\mathcal{Z}\left\{\frac{y[k] - 2y[k-1] + y[k-2]}{T_e^2} + \frac{y[k] - y[k-1]}{T_e}\right\}$ $= Y(z) - 2z^{-1}Y(z) + z^{-2}Y(z) + (\dots)$ $(\dots)$ $G_0(z) = \frac{0,7z + 0,07}{(z-0,36)(z-0,13)} \quad \text{circuit deschis}$ $Y(z) - 2z^{-1}Y(z) + z^{-2}Y(z) + (\dots)$ $G_0(z) = \frac{G(z)}{1 + G(z)} = \frac{0,7z + 0,07}{z^2 + 0,12z + 0,11}$ 2) $e^*_{st} = \frac{1}{1 + \lim_{z \rightarrow 1} G(z)} = 0,41$ $\uparrow$ K_p^* se det. pe circuit deschis dublu integrator $\Rightarrow$ $K_p^* = \lim_{z \rightarrow 1} G(z) = \frac{0,7z + 0,07}{(z-0,36)(z-0,13)}$

3) pe circuit închis

 $\Rightarrow$ sist. este stabil4) $y(kT_e) = \mathcal{Z}^{-1}\{Y(z)\} \quad R(z) = \frac{z}{z-1}$ $Y(z) = \frac{z}{z-1} \cdot \frac{0,7z + 0,07}{z^2 + 0,12z + 0,11} = \frac{0,7z^2 + 0,07z}{z^3 - 0,79z^2 - 0,17z - 0,1} \Big| \cdot z^{-3}$ $Y(z^{-1}) = \frac{0,7z^{-1} + 0,07z^{-2}}{1 - 0,79z^{-1} - 0,11z^{-2} - 0,17z^{-3}}$ $\frac{0,7z^{-1} + 0,07z^{-2}}{-0,79z^{-1} + 0,56z^{-2} + (\dots)} \Big| \frac{1 - 0,79z^{-1} - 0,11z^{-2} - 0,17z^{-3}}{0,17z^{-1} + 0,63z^{-2}}$ $= 0,63z^{-2} + (\dots)$ $y(kT_e) = \{0; 0,7; 0,63\}$ $\frac{1}{s^2 + 3s + 2} = \frac{kw_m^2}{s^2 + 2\beta w_m s + w_m^2}$ $w_m^2 = 3 \quad 2\beta w_m = 3 \quad \beta = \frac{3}{2w_m} = \frac{3\sqrt{3}}{2 \cdot 3}$ $y(kT_e)$ 0 kT_e e^*_{st}